METHOD 21 / A COMPLETE BEGINNER GUIDE

Spectral coherence

Question this method answers: At which cycle lengths (frequencies) do two FX return series oscillate together?

Spectral coherence $\gamma^2(f)$ is between 0 and 1 at each frequency f. It has no sign — phase carries timing separately.

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Teaching sinusoid returns only — no live spectral panels from snapshots.

Prepared for: 21 - Spectral Coherence. README status: not started.

02 / THE BASIC IDEA

Co-movement by frequency

Daily Pearson r mixes all cycle lengths into one number. Spectral methods split variation across frequencies f (cycles per day). Coherence asks: when X has power at f, does Y share it?

Unsigned strength

$\gamma^2(f) \in [0, 1]$	0 = no linear linkage at f; 1 = perfect linear linkage at f.
No sign	Opposite phase can still yield high γ^2 if alignment is stable.
Smoothing	Welch averaging trades bias for variance — window length matters.

03 / RETURNS ON THE CLOCK

FX percent math (majors)

```
Simple return = P_t / P_{t-1} - 1
Log return = ln(P_t / P_{t-1})
```

1.1000 → 1.1110 is about +1% simple return. Spectral coherence is applied to evenly spaced return series (D1 majors). Gaps should be handled before FFT-based steps — do not silently fill with zeros without documenting.

Example from the formula sheet: a five-day cycle corresponds to frequency f = 1/5 cycles per observation when one step equals one trading day.

04 / SXX, SYY, SXY

Spectral building blocks

Sxx(f) measures how much series X wiggles at frequency f. Syy(f) does the same for Y. Sxy(f) is the cross-spectrum linking X and Y at f — it can be complex.

Coherence normalizes $|Sxy|^2$ by both own-power spectra so γ^2 is comparable across pairs.

Think of Sxx as a volume knob per frequency for X alone. When Sxy is large at f but Sxx is tiny, γ^2 can still be moderate because shared power must be judged relative to each leg’s own activity at that f.

For FX returns, demeaning before FFT/Welch steps removes the zero-frequency spike from slow drift — state whether you demeaned or detrended in experiment NOTES.

Welch method in SciPy

scipy.signal.coherence implements Welch’s method: segment the series, window each segment, FFT, average periodograms across segments. That smoothing stabilizes $\gamma^2(f)$ for noisy FX returns.

Parameters nperseg, noverlap, and window shape change the frequency grid and bias — freeze them in experiment NOTES.

06 / SMALL MATH DICTIONARY

Symbol dictionary

Symbol	Meaning
f	Frequency (cycles per observation).
Sxx(f), Syy(f)	Auto-spectral density (power) at f.
Sxy(f)	Cross-spectral density at f.
$\gamma^2(f)$	Magnitude-squared coherence at f.
Phase $\phi(f)$	Angle of Sxy — timing offset at f.
Welch	Averaged periodogram estimator (SciPy default).

Section 21 in forex-formulas-explained.pdf uses the same γ^2 normalization.

07 / NORMALIZE SHARED POWER

Coherence formula

```
gamma^2(f) = |S_xy(f)|^2 / ( S_xx(f) * S_yy(f) )
```

At each f, divide squared cross-power by the product of auto-powers. Values above 1 indicate a numerical or scaling bug — correct implementations clamp to [0, 1].

$\gamma^2(f)$ describes linear co-sinusoidal linkage at f, not nonlinear tail dependence.

08 / SHARED FIVE-DAY RHYTHM

Teaching sinusoids

Made-up returns follow sin waves with shared period 1/0.2 = 5 days on D1 steps. Amplitudes differ (0.8% versus 0.6%) but phase offset is fixed.

```
Welch peak coherence ≈ 1.0000
Peak frequency ≈ 0.2500 cycles/day
```

Peak location should sit near the planted frequency after smoothing — small shift is normal when nperseg does not divide the period evenly.

09 / TIMING AT FREQUENCY F

Read phase separately

```
S_xy(f) = |S_xy| * exp(i * phi(f))
```

Phase $\phi(f)$ tells whether Y leads or lags X at that cycle length. Coherence alone hides that sign information — report $\phi(f)$ alongside $\gamma^2(f)$ when lead-lag matters.

For FX risk, a high γ^2 at low f (slow cycles) may reflect shared macro drift; high γ^2 at high f may reflect short-lived liquidity shocks — inspect both.

10 / REPO USAGE

Daily majors workflow

1. Evenly spaced return matrix

Align eight majors on common business days from a frozen snapshot.

2. Pre-specify Welch settings

Document nperseg before scanning all pairs.

3. Export $\gamma^2(f)$ curves, not only the max

Peaks can be spurious with short samples — show the whole curve.

Compare to Pearson and lags

Pearson r is one number; $\gamma^2(f)$ is a function. Lagged cross-correlation (method 12) stays in the time domain. Use spectral coherence when the hypothesis names a cycle length (weekly rebalancing, month-end flows).

None of these prove causation or deliver a hedge ratio.

Software reproduction

```
from scipy import signal
f, Cxy = signal.coherence(x, y, fs=1.0, nperseg=16)
# Cxy is magnitude-squared coherence (gamma^2)
```

Match fs to your sampling (1.0 for one observation per day). Detrend or demean returns if slow drift would dominate low frequencies — state the choice.

Alternative: numpy FFT periodogram without Welch is noisier; prefer signal.coherence for teaching reproducibility.

Limits

Short samples give ragged spectra. Non-stationarity (crisis regimes) violates fixed-spectrum interpretations unless you window data.

Coherence is linear in frequency space — strong nonlinear tail links may be missed.

Sharpening γ^2 by chasing peak frequency across pairs without pre-registration inflates false positives.

Practice

Q: Can $\gamma^2(f)$ be negative? **A:** No — it is squared magnitude, between 0 and 1.

Q: Does high γ^2 at f = 0.2 prove Y causes X? **A:** No — use phase and domain knowledge.

Check: teaching peak $\gamma^2 \approx 1.0000$ with nperseg=16.

Q: Is γ^2 the same as |Pearson r|? **A:** No — r mixes all frequencies; $\gamma^2(f)$ is localized.

Sources

[1] SciPy documentation: scipy.signal.coherence (Welch magnitude-squared coherence).
Implementation used in this repo guide.
Open source

[2] forex-formulas-explained.pdf — section 21 (this repository).
$\gamma^2(f)$ definition.
Open source

Rebuild: python build_guide.py. End of spectral coherence beginner guide.